Homework 5

Problem 1

求 a_n 的非递推形式
(1)$a_0=2, a_1=3, a_{n+2}=3a_n-2a_{n+1}$
(2)$a_0=1, a_{n+1}=2a_n+3$

Answer:

(1)特征方程 $x^2+2x-3=0$ 解得 $x_1=-3, x_2=1$
令 $a_n=a(-3)^n+b$

$$\begin{cases} a+b=2 \\ -3a+b=3 \end{cases} \qquad \begin{cases} a=-\dfrac{1}{4} \\ b=\dfrac{9}{4} \end{cases}$$

$\therefore a_n=-\frac{1}{4}(-3)^n+\frac{9}{4}$

(2)特征方程 $x=2$ 齐次解 $a_n'=p\cdot 2^n$ 通解 $a_n=p\cdot 2^n+q$

$$\begin{cases} p+q=1 \\ 2p+q=5 \end{cases} \qquad \begin{cases} p=4 \\ q=-3 \end{cases}$$

$\therefore a_n=4\cdot 2^n-3$

Problem 2

$a_{n+2}=\sqrt{a_{n+1}a_n}$，$a_0=2, a_1=8$，求 a_n 的解析式及 $n\to\infty$ 时的 a_n

Answer:

令 $b_n=\log_2 a_n$
则有 $\log_2 a_{n+2}=\frac{1}{2}(\log_2 a_{n+1}+\log_2 a_n)$
即 $2b_{n+2}=b_{n+1}+b_n$
特征方程 $2x^2-x-1=0$
解得 $x_1=-\frac{1}{2}$，$x_2=1$
设 $b_n=p(-\frac{1}{2})^n+q$

根据 $b_0 = 1, b_1 = 3$
有

$$\begin{cases} p + q = 1 \\ -\frac{1}{2}p + q = 3 \end{cases} \qquad \begin{cases} p = -\frac{4}{3} \\ q = \frac{7}{3} \end{cases}$$

$$\therefore b_n = -\frac{4}{3}(-\frac{1}{2})^n + \frac{7}{3},\ a_n = 2^{-\frac{4}{3}(-\frac{1}{2})^n + \frac{7}{3}},\ \lim_{n\to\infty} a_n = 2^{\frac{7}{3}}$$

Problem 3

求证：$\forall n \geq 0, \frac{1}{2}[(1+\sqrt{2})^n + (1-\sqrt{2})^n] \in \mathbb{Z}$

Answer:

$\lambda_1 = 1+\sqrt{2}, \lambda_2 = 1-\sqrt{2} \qquad \lambda_1 + \lambda_2 = 2, \lambda_1\lambda_2 = -1$
构造以 λ_1, λ_2 为根的一元二次方程 $x^2 - 2x - 1 = 0$
$n = 0, 1$ 时 $\frac{1}{2}\left[(1+\sqrt{2})^n + (1-\sqrt{2})^n\right]$ 分别为 $1, 1$
构造数列 $a_0 = 1, a_1 = 1, a_{n+2} = 2a_{n+1} + a_n$ 的解析式为 $a_n = \frac{1}{2}[(1+\sqrt{2})^n + (1-\sqrt{2})^n]$
由于只涉及正整数的加减乘运算，所以数列的各项均为整数，即 $\forall n \geq 0, \frac{1}{2}[(1+\sqrt{2})^n + (1-\sqrt{2})^n] \in \mathbb{Z}$

Problem 4

计算

$$\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} (-4)^{-k}$$

Answer:

令 $f(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} (-4)^{-k}$

$$
\begin{aligned}
f(n) &= \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} (-4)^{-k} \\
&= 1 + \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n-k}{k} (-4)^{-k} \\
&= 1 + \sum_{k=1}^{\lfloor n/2 \rfloor} \left[\binom{n-1-k}{k} + \binom{n-1-k}{k-1} \right] (-4)^{-k} \\
&= \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-1-k}{k} (-4)^{-k} + \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n-1-k}{k-1} (-4)^{-k} \\
&= \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-1-k}{k} (-4)^{-k} + \sum_{k=0}^{\lfloor (n-2)/2 \rfloor} \binom{n-2-k}{k-1} (-4)^{-(k+1)} \\
&= f(n-1) - \frac{1}{4} f(n-2)
\end{aligned}
$$

特征函数 $x^2 - x + \frac{1}{4} = 0$ 解得 $x_1 = x_2 = \frac{1}{2}$

设 $f(n) = \frac{an+b}{2^n}$

$$
\begin{cases}
f(0) = b & = 1 \\
f(1) = \dfrac{1}{2}(a+1) & = 1
\end{cases}
\qquad
\begin{cases}
a = 1 \\
b = 1
\end{cases}
$$

$\therefore f(n) = \frac{n+1}{2^n}$